FactoryBench: Evaluating Industrial Machine Understanding

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Code: `anonymous.4open.science/r/FactoryBench`
Dataset: `huggingface.co/datasets/FactoryBench/FactoryBench`

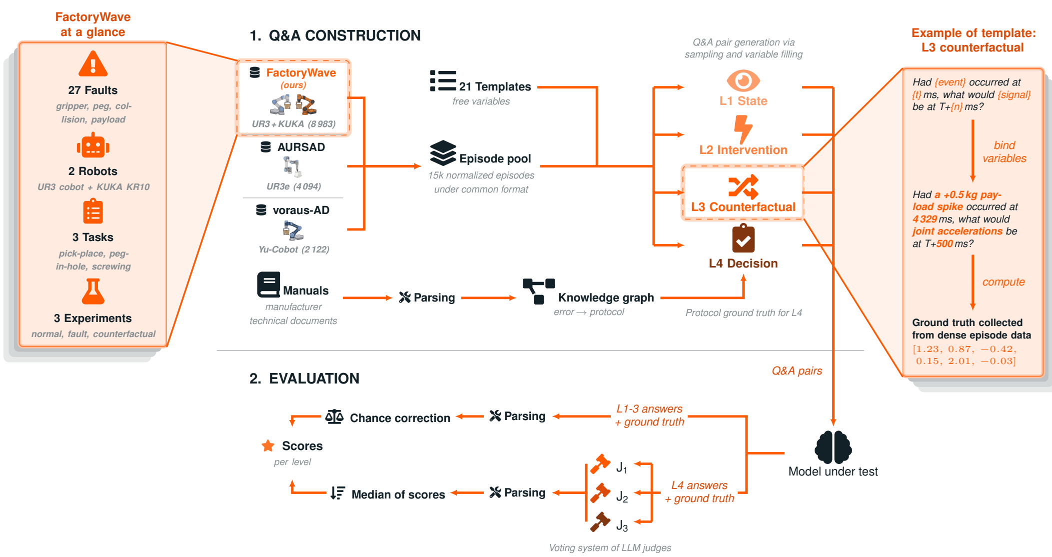

Figure 1: End-to-end FactoryBench pipeline. **(1) Q&A construction:** normalized episodes from FactoryWave, AURSAD [1], and voraus-AD [2] are paired with typed-variable templates (plus the knowledge graph for L4) to generate Q&A items at four reasoning levels. **(2) Evaluation:** the model's answers are scored deterministically for L1–3 and by the median of three LLM judges for L4 free-form, then chance-corrected per level.

Abstract

Robot foundation models are increasingly deployed **zero-shot**: downloaded pre-trained and pointed at a machine they were never trained on. We ask what such a model understands about a physical system it is seeing for the first time, and introduce **FactoryBench**, a zero-shot benchmark over dense industrial robot telemetry. Q&A pairs span four levels (state estimation, intervention, counterfactual reasoning, and engineering decision-making) and five answer formats, four scored deterministically and free-form answers by an LLM-as-judge protocol. We release **FactoryWave** (a dense multitask sensor dataset from a UR3 cobot and a KUKA KR10 industrial arm) and build FactoryBench as over 70k items grounded in roughly 15k episodes from FactoryWave, AURSAD [1], and voraus-AD [2]. Across six frontier models, none exceeds 50% (chance-corrected) on the structured levels, two fail to beat a linear-regression baseline on state estimation, and the ranking reshuffles entirely on decision-making, with the leader on the first three levels falling to the bottom of the fourth. A human expert reaches 0.96–0.98 on

the same items, so the headroom is real rather than an artifact of unanswerable questions. Zero-shot transfer to an unseen industrial machine is not yet reliable enough to act on.

1 Introduction

The premise of zero-shot Physical AI is that a pretrained model can be pointed at a machine it has never seen and be immediately useful, with no task-specific data collection or finetuning on that platform. Industrial settings are where this would pay off most and where it is hardest to validate: production robots are expensive to instrument and cannot be run to failure for data collection. If a generalist model could diagnose an unfamiliar arm from its signals alone, the economics of industrial monitoring would change. Whether it can is underdetermined, because the evaluations that would answer it do not exist.

Zero-shot competence on a physical system is not one capability but several: reading operational state from raw multivariate signals, predicting what an intervention does to that state, reasoning counterfactually about histories that did not occur, and turning all of it into an action an operator could execute. These come apart, since a system can track a signal well and still have no idea what to do about it, and existing evaluations do not separate them. General benchmarks such as MMLU, BIG-bench, and MMMU [3, 4, 5] probe textual or multimodal understanding rather than machine behavior, and time-series Q&A benchmarks are typically synthetic or univariate and omit counterfactuals. Neither specialized time-series models [6, 7, 8], which generalize poorly off-task [9, 10, 11], nor LLMs on dense numerical series [12, 13], have been evaluated on the full set.

Evaluating on telemetry alone matches where industry is heading. Plants increasingly run *lights-out* to cut labour and energy costs: FANUC has operated a lights-out robot factory since 2001, running unsupervised for up to 30 days with lighting and HVAC reduced or eliminated [14], and Xiaomi's Changping facility builds on the order of 10 million smartphones a year the same way [15]. The human who would have caught a fault by sight or sound is off the floor, and where cameras are deployed it is overwhelmingly for *product* inspection rather than machine health [15], which lives in quantities no camera observes: joint torque, motor current, tracking error, controller mode. Controller telemetry is emitted by default by every industrial machine and is the one modality guaranteed across a heterogeneous plant.

We evaluate LLMs and LLM-driven agents rather than specialised classifiers because in an unattended plant the artefact that matters is the justification, not the label. A classifier emits a fault class with no account of which signals produced it; a language model emits a trace from signal to root cause to procedure, checkable against the telemetry and the vendor manual, and rejectable when the reasoning is unsound even if the label is right. Downtime sets the stakes: unplanned downtime costs Fortune Global 500 industrial organizations almost \$1.5 trillion a year, 11% of turnover, and an idle automotive line runs past \$2 million an hour [16].

Testing language models is also an increasingly good proxy for testing robot policies: current vision-language-action models are largely built by post-training a pretrained LLM or VLM backbone rather than training a controller from scratch (RT-2 [17], OpenVLA [18], π_0 [19]), so the reasoning limits we measure here bound what the resulting VLA can do on an unfamiliar platform, and are cheaper to measure than full policy rollouts.

We propose FactoryBench to measure exactly this, evaluating every model strictly zero-shot: no finetuning, no in-context examples from the target platform, no exposure to the robot under test. It is grounded in **FactoryWave**, a dense multivariate dataset from collaborative and industrial one-arm platforms (UR3 at 125 Hz, KUKA KR10 at 83 Hz) with synchronized setpoint, context, feedback, and effort signals. To our knowledge it is the first extensive anomaly dataset to (partly) cover an industrial robot, which poses the cross-embodiment question directly: the cheap-to-instrument cobot and the production-grade industrial arm appear under the same schema. A framework of 21 structured templates, parameterized over generic primitives rather than hard-coded to a machine, instantiates over FactoryWave (and AURSAD / voraus-AD) to yield the full corpus, and extends to a new platform by populating its signal mapping.

2 Related work

Recent advances in LLM reasoning include prompting and deliberative strategies (e.g., Chain-of-Thought and Tree-of-Thought) [20, 21], alongside tool-augmented paradigms such as Toolformer and ReAct [22, 23, 24]. Time-series modeling has advanced through transformer-based architectures (Informer, Autoformer, FEDformer, PatchTST) [6, 25, 26, 27], strong alternative baselines such as TFT and N-BEATS [28, 29], and critical comparisons against simpler linear models [11]. Foundation-model directions (e.g., Chronos) explore transfer and zero/few-shot adaptation [30, 31], with large-scale industrial corpora such as FactoryNet [32], on which FactoryBench builds. Anomaly and reliability monitoring literature includes OmniAnomaly, USAD, and Deep SVDD [7, 8, 33], with surveys highlighting persistent gaps in interpretability and root-cause actionability [9, 10].

Causal frameworks provide formal tools for interventions and counterfactuals [34, 35, 36], while temporal methods such as Granger causality and nonlinear causal discovery support directional reasoning in time series [37, 38]. Generalist robot policies and robot foundation models are typically evaluated on task success under visuomotor control; FactoryBench is complementary, holding the platform fixed and asking what a model understands about a machine it has never seen, from the controller and sensor signals that any deployed system already emits, which is the modality available in industrial settings where cameras and teleoperation demonstrations are not. Against the closest Q&A and time-series benchmarks (TimeSeriesExam [39], ChatTS [40], EngineMT-QA [41], Time-MQA [42], QuAnTS [43], CLEVRER), FactoryBench is the only one that is simultaneously robot-grounded, multivariate, counterfactual, and accompanied by a new dense high-frequency dataset rather than relying exclusively on public or synthetic sources.

3 FactoryBench framework

3.1 Four levels of machine understanding

Our four-tier hierarchy builds on Pearl's ladder of causation association, intervention, and counterfactual reasoning, the organizing principle for causal inference and an increasingly common evaluation axis for machine-learning systems [34, 35, 44, 45, 46]. We add a fourth decision-making level reflecting how industrial robotic platforms are actually operated: after interpreting state and causal relationships, operators must select and execute corrective procedures, typically prescribed by vendor manuals. This aligns with the diagnose-then-act loop standard in fault-tolerant control, and grounds each level in a qualitatively distinct reasoning skill with interpretable failure modes.

Level 1: State. Assesses the agent's ability to interpret the current state of the machine under normal conditions, including identification of operational modes and recognition of sensor patterns.

Level 2: Intervention. The agent must reason about the consequences of interventions or events occurring at the present timestep, including predicting the immediate impact of control actions and diagnosing emerging faults in real time.

Level 3: Counterfactual. Evaluates reasoning about hypothetical scenarios, requiring the agent to simulate alternative histories and assess how outcomes would differ under counterfactual conditions.

Level 4: Decision Making. The highest level evaluates the agent's ability to act on its understanding of the system: given a recorded episode and a task description, the agent must produce a diagnosis or remediation grounded in the previous three levels, reflecting what we expect from an LLM acting as an engineering assistant on the factory floor.

Each question is emitted in one of five answer formats (single-select MCQ, multi-select MCQ, ranking, tensor, free-form); the four structured formats are scored deterministically and free-form answers are scored by an LLM-as-judge voting protocol.

3.2 Time series data and causal schema (SCE)

So that the dataset structure itself encodes causality, all time-series signals are organized into three causal groups: **Setpoint** (the controller's commanded target), **Context** (physical and configured conditions), and **Effort+Feedback** (the machine's measured response). Under healthy operation the response is a known function of setpoint and context; faults manifest as structured residuals from that baseline, giving a single operational fault definition that applies across machines and tasks.

The data conforms to this schema across two source types: **FactoryWave**, a custom dataset generated from one-arm robotic platforms executing industrial tasks under varied conditions and systematically injected anomalies, and **open-source datasets** such as AURSAD [1] and voraus-AD [2], preprocessed to conform to the unified episode structure. All three sources provide the full setpoint–context–effort triplet at 83 Hz or higher (Table 1).

Table 1: Datasets incorporated into FactoryBench. FactoryWave is collected and released with this work; the other two are open-source datasets adapted to our schema. Tasks: PnP = pick-and-place, Scr = screwing, PiH = peg-in-hole.

Dataset	Robot	Episodes	Frequency	Tasks	Anomalies
AURSAD [1]	UR3e	4094	100 Hz	Scr	4
voraus-AD [2]	Yu-Cobot	2122	100/500 Hz	PnP	12
FactoryWave (ours)	UR3, KUKA KR10	8983	125/83 Hz	PnP, Scr, PiH	27

4 Reliable Q&A generation

4.1 Scalable generation via extensive labeling

Scalable ground-truth generation is the central challenge of any Q&A benchmark grounded in raw data. FactoryBench couples a structured labeling ontology with a context-free-grammar-style template system: each of the 21 templates contains variable slots filled at generation time either from label information read directly from the episode (signal names, timestamps, anomaly labels, root causes) or from a predefined sampling distribution attached to that slot (prediction horizons, thresholds, curated vocabularies). Answer options for single- and multi-select questions are drawn from a shared pool of pre-defined verifiable statements, each paired with a rule evaluated deterministically against the time series, so ground truth is never imputed but computed directly from the underlying labels; for single-select MCQ, the sampling step enforces that exactly one of the four options evaluates to true, so the question is unambiguous by construction. This is also our primary defense against template-generated benchmarks being solvable from surface structure alone: every template admits a combinatorially large instantiation space over signal names, timestamps, prediction horizons, and injected-fault types, so the correct answer is never recoverable from the question text alone and is always determined by the paired time series, which varies across instantiations. Alongside the full corpus we release **FactoryBench-Lite**, a 3,000-item subset balanced across all 21 templates and, within each template, on the dimension that determines its answer, so that a cheap evaluation is not weighted by how many items a template happens to contribute.

4.2 Density of FactoryWave

FactoryWave is collected from two physical robots (UR3 and the KUKA KR10) executing up to three industrial tasks each: pick-and-place, screwing, and peg-in-hole. Each complete task cycle constitutes one episode, recorded at 125 and 83 Hz across over 100 sensor channels covering joint setpoints, feedback, torques, speeds, estimated contact forces, TCP pose, gripper state, and task-phase labels. To compensate for the limited tool-side telemetry exposed by the KUKA KSS controller, we additionally mount a 9-DoF inertial measurement unit on the KUKA gripper, streamed in sync with the controller signals.

Episodes are organized into three experiment types. **Normal** episodes capture nominal operation across three payload levels. **Fault** episodes inject one of 27 fault types (gripper failures, misconfigurations, collisions, and peg-in-hole-specific anomalies), each with fault-specific metadata and, where applicable, a precise injection timestep. **Counterfactual** episodes provide ground truth for Level 3 reasoning by approximating the do-operation on physical hardware: for each baseline we re-execute the task 3–5 times with every controllable condition held fixed and inject the target fault at a fixed timestep t, then keep the run whose pre-injection segment minimizes the signature kernel MMD against the baseline, using it as an approximate ground truth for hypothetical divergence.

4.3 Knowledge graph and protocol extraction

Reliable ground truth for Level 4 troubleshooting requires machine-specific recovery protocols grounded in manufacturer documentation. For each robot we parse the manufacturer's official runtime error documentation into a structured mapping from error codes to descriptions and recommended recovery steps, then map every anomaly type in FactoryBench to the most likely corresponding manufacturer error code, with PhD-level robotics experts reasoning about the physical cause of each anomaly. Where a physical fault injection has no well-defined controller error, the same experts author the recovery protocol directly. The complete mapping is ingested into a knowledge graph alongside robot specifications, task definitions, and signal schema, which the generation pipeline queries at question-generation time to inject ground-truth protocol context into Level 4 items without per-question human review.

5 Evaluation and results

5.1 Zero-shot evaluation

We evaluate a cross-vendor panel of frontier LLMs available at submission time (April 2026): Claude Sonnet 4.6 [47] and GPT-5.1 [48] on the closed-weight side, and DeepSeek V3.2 [49], Mistral Large 3 [50], and Qwen3-235B [51] on the open-weight side, plus Qwen3-4B [51] as a lightweight reference. For calibration we report a non-LLM baseline that dispatches per item by answer format: linear regression on the target signal for scalar and tensor templates, and uniform random over the option set for single- and multi-select MCQ and ranking templates (free-form Level 4 templates are excluded, as no traditional method maps cleanly to producing a remediation protocol). Figure 2 reports chance-corrected accuracy for the full panel.

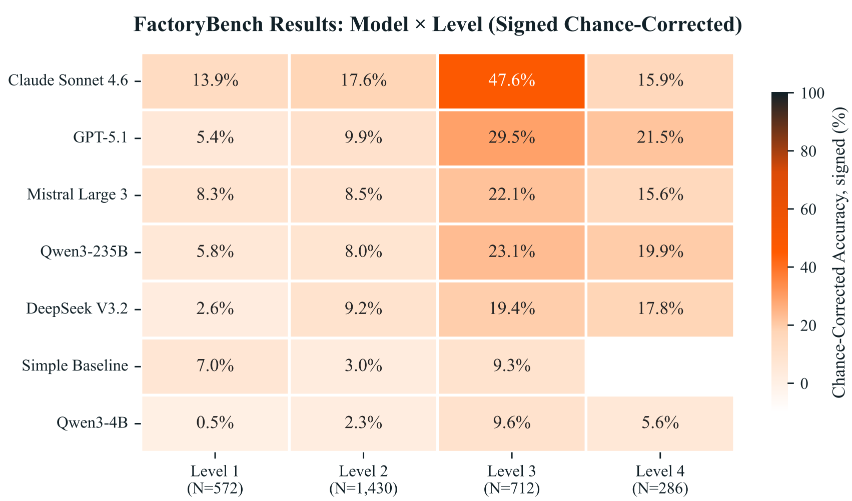

Figure 2: Main results: chance-corrected accuracy (%) for the cross-model panel on FactoryBench's four reasoning levels.

5.2 Signal comprehension across Levels 1–3

Most models fail to separate from the simple baseline on Level 1, including the largest in the panel. Under the signed chance correction the non-LLM baseline lands at $+4.3\%$ on L1. Four of six LLMs fail to clear it: Qwen3-4B (-1.8%) and DeepSeek V3.2 (-0.3%) score below chance outright, and GPT-5.1 ($+3.2\%$) and Qwen3-235B ($+3.6\%$) sit below the baseline despite being the largest models in the panel. Only Mistral Large 3 ($+5.9\%$) and Claude Sonnet 4.6 ($+11.8\%$) clear it, and Mistral only barely. Raw scale does not reliably confer the ability to extract precise state from dense industrial signals.

Models that struggle on Level 1 still outperform the baseline on Levels 2 and 3, and rank order is stable across the three. DeepSeek V3.2 ($+5.8\%$, $+17.1\%$) and GPT-5.1 ($+6.9\%$, $+27.5\%$) both lift clear of the L2/L3 baseline values (-0.2%, $+6.2\%$) despite trailing it on L1, yet the model rank order is nearly identical across L1–L3: Claude Sonnet 4.6 leads at L2 and L3 ($+14.6\%$, $+45.7\%$), Qwen3-235B is the strongest open-weight model ($+20.6\%$ on L3), and Mistral Large 3 follows ($+5.3\%$, $+19.7\%$). This points toward state comprehension propagating into intervention and counterfactual reasoning rather than being substituted by it.

A human expert reaches near-ceiling on Levels 1–2, but not on Level 3. A robotics expert answered a 60-item sample under open-tool, no-LLM conditions, graded by the benchmark's own scorer, reaching 0.962 at L1 and 0.980 at L2 against 0.712 and 0.662 for the strongest model: the gap is not an artifact of ambiguous items, and reading the signal out is what the panel fails at. At L3 the expert reaches only 0.625 and does not lead the panel, which we attribute to counterfactual prediction being poorly served by existing tooling rather than to defective items.

5.3 Engineering decision-making at Level 4

Level 4 exposes an industrial readiness gap, and the panel ordering reshuffles completely. On Level 4 (uncorrected 0/0.5/1 rubric, not directly comparable to the chance-corrected L1–L3 scores) the leader reaches 21.5% and the other five score between 5.6% and 19.9%. Claude Sonnet 4.6, which leads L1–L3 and more than doubles the next model on L3, falls to fourth of six on L4 at 15.9%. GPT-5.1 leads L4 at 21.5%, and Qwen3-235B (19.9%) and DeepSeek V3.2 (17.8%) also overtake Claude Sonnet 4.6 there. The dissociation between *signal comprehension* and *protocol retrieval* means optimizing for one does not deliver the other: even models with strong L1–L3 comprehension either lack the machine-specific knowledge to diagnose faults, or cannot ground that knowledge in the observed signals, and closing this gap is the central challenge FactoryBench is designed to measure.

The L4 collapse is partly a read-out failure, not only a representational one. We probe this directly: linear probes on a frozen open-weight panel member recover fault *presence* from the model's activations well above chance (0.83) even where the model's own answer stays below chance, whereas fault *type* is only weakly decodable and no better than an untrained-network baseline. The information a correct L4 answer needs is thus partly already in the model's representations; part of what fails is turning it into a verbalized diagnosis, not only building it in the first place.

5.4 Limitations

FactoryBench has three substantive limitations. Level 3 *counterfactual* ground truth on hardware is approximate: two real runs cannot share an identical pre-injection state, so our signature-kernel-MMD-selected pair proxies the do-operation, conflating model reasoning with proxy quality. Our fault model is simplified: atomic, fast faults from a closed catalogue of 27 mechanisms rather than the compound, gradual faults of production, closer to in-distribution fault *recognition* than *detection*. Level 4 ground truth comes from a finite catalogue of manufacturer codes and expert-authored procedures, so a model is partly evaluated on closed-book retrieval it may not have seen in pretraining.

6 Conclusion

We introduced FactoryBench, a four-level benchmark for machine understanding over industrial robotic telemetry, grounded in FactoryWave and structured around Pearl's causal hierarchy. Zero-shot evaluation of six frontier LLMs shows that signal comprehension and protocol-grounded decision-making do not co-develop: the L1–L3 ordering reshuffles entirely on L4, and structured-level scores stay well below 50% (chance-corrected). Zero-shot deployment onto unseen hardware is not yet reliable: models transfer some competence to a machine they have never seen, but not enough for an operator to act on it, at the level that matters most. Extended results and protocols are released with the dataset and code (links above).

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